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Polysaccharide epitopes recognized by human α-L-Rha antibodies

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ABSTRACT

The levels of antibodies in human blood binding to the monosaccharide α-L-Rha are the highest among all anti-glycan antibodies. Moreover, anti-Rha antibodies are found in all individuals, suggesting that they are naturally occurring rather than adaptive immunoglobulins. Rhamnose is common in both bacterial (especially infectious) and plant polysaccharides, however, it remains poorly understood which rhamnose-containing epitope(s) – whether monosaccharide, oligosaccharide, or complex molecular patterns – are recognized by human antibodies. Using an affinity adsorbent, α-L-Rha-Sepharose, antibodies were isolated from human immunoglobulin preparations (IgG + IgM + IgA) and analyzed using highly representative arrays of bacterial and plant polysaccharides (about 1000 glycans, of which >240 contained Rha). Isolated anti-α-L-Rha antibodies bound to almost all polysaccharides where the rhamnose residue was located either terminally or as α1–2, 1–3 or 1–4 linked pendant substituent, but not to internal positions, regardless of whether they were bacterial *O*-antigens or plant polysaccharides. It was concluded that human polyclonal anti-α-L-Rha antibodies have a reasonably narrow range of epitope specificity. The recognition of a small-sized monosaccharide epitope, on one hand, and a high proportion of IgM and IgA, on the other, suggest a high degree of polyvalence in recognizing the natural targets of the antibodies studied. *In vivo*, anti-α-L-Rha antibodies are more likely to recognize a pattern composed of tightly packed lipopolysaccharides (or capsular polysaccharides, or plant cell walls) rather than repetitive epitopes on a single polysaccharide molecule.

1. Introduction

Key players in the humoral anti-bacterial response are antibodies (Abs) to polysaccharides (PS), including both adaptive [1,2] and innate (naturally occurring) [3–5]. The repertoire of the latter is quite broad [6] and appears to be uniform when studying large cohorts of healthy donors. In most donors, the highest levels of Abs (when considering both titer and affinity) belong to anti-L-Rha immunoglobulins, i.e., Abs that bind the monosaccharide α-L-rhamnopyranose [7]. When sera of individuals were tested using a printed glycan array (PGA), all of them were found to have high-to-moderate levels of α-L-Rha Abs [8,9]. This is not surprising, as rhamnose is widely distributed in nature [10], being a common component of various bacterial and plant polysaccharides as well as plant glycosides ([11–15], http://csdb.glycoscience.ru/database). Additionally, rhamnose is also found in viruses [16], fungi [17], and lower animals [18]. The high levels of these Abs have sparked

Abbreviations: CIP, complex immunoglobulin preparation; PGA, printed glycan array; EIA, enzyme immunoassay; PS, polysaccharides; Abs, antibodies; nAbs, naturally occurring antibodies.

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interest in using rhamnose antigens as an anti-tumor therapeutic. The lipophilic glycoconjugate introduced into the tumor is integrated into the plasma membrane of the tumor cells, initiating a systemic immune cascade due to the recruitment of existing (naturally occurring) antibodies to rhamnose, which suggests it may become an effective approach to oncotherapy [19–21].

The binding epitope of anti-Rha Abs, i.e., their specific target fragments in bacterial or plant polysaccharides, remains unknown. Moreover, there is no evidence indicating whether the primary target is bacterial *O*-antigen, capsular PS, plant PS, or another glycoconjugate. Notably, there is a high structural diversity among rhamno-PS epitopes. The Rha fragment can be found in both internal and side (pendant) positions of PS chains. Specifically, rhamnose can link to a penultimate monosaccharide residue through 1–2, 1–3, 1–4, or 1–6 bonds, and can be present not only as α-L-, but also, albeit rarely, as β-, and D-glycopyranose (for example [22], and http://csdb.glycoscience.ru/database). Since some naturally occurring Abs (nAbs) are capable of binding to internal glyco-epitopes [23], the possibility of recognizing the internal regions of the polysaccharide chain should also be considered for anti-α-L-Rha Abs. And a number of other questions were supposed to be answered. Namely, are anti-rhamnose immunoglobulins naturally occurring or adaptive, and if natural, what is their primary target? To answer these questions, human antibodies were isolated by affinity chromatography on an α-L-Rha monosaccharide adsorbent, and their specificity was studied by several methods, including a bacterial PS array (334 polysaccharides), a combined PS array (bacterial PS and oligosaccharides, a total of 782 ligands, of which more than 100 contained Rha), and a plant PS array (212 ligands, of which 142 contained more than 1 mol.% Rha), for a total of about 1000 glycoligands.

2. Materials and methods

2.1. Reagents

Complex immunoglobulin preparation (CIP), which contains IgG, IgM, and IgA, and intramuscular immunoglobulin G were from Microgen (Moscow, Russia). Fluorescently labeled secondary antibodies, including goat anti-human IgG-Alexa^{555} and goat anti-human IgM-Alexa^{647}, were sourced from Invitrogen (Eugene, OR, USA), while goat anti-human IgA-Cy3 was acquired from Jackson ImmunoResearch (Ely, Cambridgeshire, UK). Unlabeled and labeled with horseradish peroxidase (HRP) goat anti-human IgG, IgM, and IgA were from Southern Biotechnology Associates, Inc. (Birmingham, AL, USA). All other chemicals, including phosphate-buffered saline (PBS), tris(hydroxymethyl) aminomethane (Tris-OH), bovine serum albumin (BSA), Tween-20, *o*-phenylenediamine, NaN_3, and H_2O_2 were obtained from Sigma (St. Louis, MO, USA), unless otherwise noted. synthetic oligosaccharides (>95 % purity), Glyc-PAA (20 kDa, 20 mol % of glycan) conjugates (Glyc – glycan residue) and PAA (poly[N-(2-hydroxyethyl) acrylamide], 20 kDa), used as negative control, were sourced from GlycoNZ (Auckland, New Zealand). The spacer glycoside α-L-Rha-O$(CH_2)_3NH_2$ was obtained by glycosylation of 3-trifluoroacetamidopropanol with rhamnose peracetate in the presence of $BF_3{\cdot}Et_2O$. Its structure was confirmed by ^{13}C and ^{1}H NMR spectroscopy.

2.2. Affinity adsorbent α-L-Rha-Sepharose

To a solution of α-L-Rha-O$(CH_2)_3NH_2$ (2.65 mg, 12 μmol) in dry DMSO (400 μL) a solution of poly(4-nitrophenyl acrylate) (580 μL of 20 mg/mL) in dry DMSO and Et_3N (84 μmol, 6.1 μL) were added, and kept for 24 h at 40 °C. This mixture was added to 20 mL of amino Sepharose 6FF in dry DMSO (total volume 30 mL), to the stirred suspension Et_3N (~30 μL) was added, and the suspension was gently agitated (orbital motion shaker) for 5 h at room temperature. After 2 h, free glycoconjugate was not observed in the supernatant according to TLC (TLC data: silica gel plates, eluent 2-propanol/MeCN/water 4 : 3 : 2, visualization - charring after treatment with 7 % H_3PO_4). Extra 4-nitrophenyl groups were quenched with ethanolamine (600 μL), the suspension was gently agitated for 2 h at room temperature. The obtained material was thoroughly washed on a filter with 25 % ethanol, water and suspended in 0.5 M $NaHCO_3$. Acetic anhydride (2 mL) was added and the suspension was gently shaken for 1 h. The adsorbent was thoroughly washed with water and 25 % ethanol. Obtained material was stored in 25 % ethanol in water at 4 °C.

2.3. Affinity isolation of human anti-α-L-Rha antibodies

Lyophilized CIP containing 300 mg of IgG + IgM + IgA was dissolved in 50 mL of PBS containing 0.02 % NaN_3 (PBS-Az). First, antibodies against the affinity matrix were removed by applying 50 mL of the CIP solution onto a column containing 3 mL of PAA-Sepharose 6FF, which was equilibrated with PBS-Az at 25 °C and a flow rate of 0.4 mL/min. The anti-matrix antibody-free material that passed through the PAA-Sepharose 6FF was then adsorbed onto α-L-Rha-Sepharose 6FF, which had been pre-washed and equilibrated with PBS-Az. The column was washed of unbound antibodies with five column volumes of PBS-Az containing 0.1 % Tween-20, followed by ten column volumes of PBS-Az until the wash no longer absorbed at 280 nm (UV-detector, Bio-Rad, Philadelphia, PA, USA). Bound anti-α-L-Rha Abs were eluted with Tris-OH (0.2 M, pH 10.4) containing 0.02 % NaN_3 at a flow rate of 0.2 mL/min, with immediate neutralization using 2 M glycine-HCl, pH 2.5, to achieve a final pH of 7.3–7.5. The Abs concentration was measured by absorbance at 280 nm (assuming 1 % Ig absorbance at 280 nm to be 14.0 absorption units) using a spectrophotometer (Ultrospec 3100 pro, Amersham, Buckinghamshire, UK) and was additionally confirmed by enzyme immunoassay (EIA) (see section 2.4.).

Adsorbents were regenerated with 0.1 M NaOH under UV control before reuse, and for long-term storage, adsorbents were washed with distilled water and stored in 25 % ethanol.

2.4. Immunoglobulin isotyping and EIA Ig quantification

MaxiSorp 96-well microtiter plates (Nunc, Roskilde, Denmark) were coated with 5 μg/mL (60 μL per well) of unlabeled anti-human IgG, anti-human IgM, or anti-human IgA in a sodium carbonate buffer (50 mM $Na_2CO_3/NaHCO_3$, pH 9.6) for 60 min at 37 °C and then washed. The plates were blocked with 1 % BSA in PBS (PBS/1 % BSA), 60 μL per well, for 45 min at 37 °C. Serially twofold diluted anti-α-L-Rha antibody or pure human reference IgG, IgM, and IgA (used for calibration) were added, 50 μL per well, incubated for 1 h at 37 °C, and then washed. HRP-labeled anti-human IgG (diluted 1 : 12000 with PBS/0.3 % BSA), anti-human IgM (diluted 1 : 4500 with PBS/0.3 % BSA), or anti-human IgA (diluted 1:6000 with PBS/0.3 % BSA) secondary antibodies were added, 50 μL per well, and incubated for 60 min at 37 °C, followed by washing. Color was developed by a 20-min incubation at room temperature in 0.1 M sodium phosphate/0.1 M citrate buffer containing 0.04 % *o*-phenylenediamine and 0.03 % of H_2O_2, 100 μL per well. The color reaction was stopped by the addition of 30 μL of 1 M H_2SO_4. The absorbance was read at 492 nm with a plate reader (Wallac1420 Multilabel Counter, Victor^2, PerkinElmer Life Sciences, Finland). Between each stage, plates were washed with PBS containing 0.1 % Tween-20 four times. All tests were performed at least in duplicate. The concentration (in μg/mL) of each isotype was determined using the corresponding calibration curve.

2.5. Inhibition analysis

Microtiter plates were coated with α-L-Rha-PAA or GlcNAcα-PAA at a concentration of 10 μg/mL (100 μL per well) in sodium carbonate buffer (50 mM $Na_2CO_3/NaHCO_3$, pH 9.6) for 60 min at 37 °C. The plates were blocked as described above. Serially diluted unconjugated spacered monosaccharides (alpha-rhamnoside and alpha-glycoside of GlcNAc) or aqueous potato sprout extract or PBS containing 0.3 % BSA as a negative

control - were added together with an equal volume of anti-α-L-Rha or anti-GlcNAcα (as control, these antibodies were affinity purified in the same way as anti-rhamnose antibodies, with the exception that the adsorbent used was GlcNAcα-Sepharose 6FF). Abs solution (final concentrations of 0.2 μg/mL), totaling 100 μL per well. The plates were incubated for 60 min at 37 °C and then washed. HRP-labeled anti-human IgM + IgG + IgA secondary antibodies, diluted 1 : 10000 with 0.3 % BSA in PBS, were added at 100 μL per well. The plates were incubated for another 60 min at 37 °C, washed, and developed as described in section 2.4. The results were presented as the percentage of inhibition against the control binding of antibodies in the absence of the inhibitors. All tests were performed at least in duplicate; the differences between the replicates did not exceed 5 %.

2.6. PGA analysis

Three types of glycoarrays (hereafter referred to as arrays) were used.

1. Oligosaccharide and bacterial PS arrays contained 481 synthetic oligosaccharides and 301 bacterial PS from the collection of Zelinsky Institute of Organic Chemistry, Moscow, Russia. Structures of these PS, along with NMR data and related references, are available at http://csdb.glycoscience.ru/bacterial. A full list of ligands is presented in Supplementary Material 1. The oligosaccharides at a concentration of 20 μM and the bacterial PS at 2 μg/mL were printed onto NHS-activated Slides H (Schott Nexterion, Jena, Germany) as six replicates, as described in Ref. [24].
2. Plant PS arrays contained 183 plant PS (isolation and purification of plant PS is described in Ref. [25]) and 29 synthetic oligosaccharides. The oligosaccharides at a concentration of 20 μM and the PS at 100 μg/mL were printed onto epoxy-activated Semiglass-Epoxy slides (Semiotik LLC, Moscow, Russia) as twelve replicates, as described in Ref. [25]. A full list of ligands is presented in Supplementary Material 2.
3. Bacterial PS arrays contained 27 synthetic oligosaccharides and 307 bacterial PS from the collection of Zelinsky Institute of Organic Chemistry, Moscow, Russia. A full list of ligands is presented in Supplementary Material 3. The oligosaccharides at a concentration of 50 μM and the PS at 10 μg/mL were printed onto NHS-activated Slides H at six replicates.

Printing was performed using a non-contact piezo-arrayer SciFlexArrayer S5 (Scienion, Berlin, Germany) with a drop volume of ~0.9 nL. Ligand immobilization on the activated slides was checked by a quality control system integrated into the printer, and the quality of immobilization was monitored using a commercial set of biotinylated lectins, poly- and monoclonal antibodies (data not shown).

Chips were blocked in 100 mM borate buffer containing 25 mM ethanolamine and 0.1 % Tween-20 (for NHS-slides) and in 150 mM borate buffer contained 50 mM ethanolamine and 0.1 % of Tween-20 (for epoxy-slides) for 2 h at room temperature. Isolated anti-α-L-Rha Abs (5 μg/mL) in PBS containing 1 % BSA and 0.1 % Tween-20 (PBS/1 % BSA/0.1 % Tween-20) were applied to the chip and incubated in a humidified chamber for 1 h at 37 °C. The chip was washed with PBS containing 0.05 % Tween-20 (PBS/0.05 % Tween-20) and incubated with a mixture of goat anti-human IgG and IgM labeled with Alexa^{555} and Alexa^{647}, respectively (both at 8 μg/mL), or goat anti-human IgA labeled with Cy3 (2 μg/mL) in PBS/1 % BSA/0.1 % Tween-20 for 1 h at 37°C. After washing with PBS/0.05 % Tween-20 and deionized water, the chip was scanned by Innoscan1100 AL fluorescence scanner (Innopsys, France) at a resolution of 10 μm. Images were processed using the ScanArray Express 4.0 (PerkinElmer, Waltham, MA, USA) with a fixed circle method and then analyzed using Microsoft Excel software. Signals were measured as the median of relative fluorescent units (RFU) within replicates with median absolute deviation (MAD) normalized to the maximal signal on the chip, expressed as 100 % (RFU_{norm}(%)). Ligands with an intensity exceeding 10 % were considered significant.

2.7. Interaction of anti-α-L-Rha antibodies with a component of potato sprout extract by EIA

Hot distilled water (95 °C, 1 mL) was added to 1 g of crushed potato sprouts, followed by incubation at 40 °C for 10 h. After that, the solid fraction was precipitated by centrifugation at 4500×*g*. The supernatant as a competitive agent was used in an inhibitory EIA (see Section 2.5 of the Material and Methods) in a series of dilutions. The following steps are described in Section 2.5 of the Material and Methods.

3. Results

3.1. Binding of isolated anti-α-L-Rha Abs to bacterial polysaccharides: PGA data

Glycoadsorbent - α-L-Rha-Sepharose - was obtained by coupling of a poly(4-nitrophenyl acrylate) conjugate of l-Rha with amino groups of amino Sepharose 6FF. Residual 4-nitrophenyl esters of the polymer were transformed to amides by treatment with ethanolamine. After washing residual amino groups of amino Sepharose were acetylated with acetic anhydride in aqueous sodium bicarbonate. The adsorbent with l-Rha contents of 0.6 μmol per 1 mL of Sepharose gel was obtained.

Immunoglobulins were isolated from CIP using α-L-Rha-Sepharose. Weight ratio of IgG : IgM : IgA = 1.0 : 3.3 : 2.8 (see section 2: Material and Methods). Anti-α-L-Rha Abs were able to bind to rhamnose-containing polysaccharides (rhamno-PS). The content of these Abs in the CIP varied slightly from batch to batch, averaging 0.06 wt%, which is comparable to other similarly isolated anti-glycan Abs [3].

Data from the oligosaccharide and bacterial PS arrays (Fig. 1; full data presented in Supplementary Material 1; for simplicity, here and elsewhere linkage types in parenthesis were not highlighted) indicate that only five of the ~100 rhamnose-containing bacterial PS, all with an unsubstituted rhamnose residue, interact with anti-α-L-Rha Abs (or anti-Rha Abs as equivalents). These include PS from two *Acinetobacter baumannii* strains, two *Escherichia coli* strains, and one *Shigella boydii* strain. As shown for these antibody-positive PS (Fig. 1), anti-Rha Abs recognized pendant residues, both as a single Rha residue (i.e., in close proximity to the backbone) and as Rha-terminated pendant oligosaccharides, where the linkage between the rhamnose residue and the penultimate monosaccharide is of two types, α1–3 or α1–4; the array used did not contain 1–6 variants (there were only polysaccharides with an acetylated Rhaα1-6 residue).

It is noteworthy that when rhamnose is an inner moiety, anti-α-L-Rha Abs did not interact with the PS. Several rhamnose-terminated PS listed below did not interact (or interacted at low levels) with the anti-α-L-Rha Abs (the non-reactive PS are presented in Table 1). Their structural similarity to each other or differences from antibody-positive PS were not obvious. Rha as a ramification of a core oligosaccharide (for example, in *Pseudomonas aeruginosa* O14 (http://csdb.glycoscience.ru/bacterial)) is silent as well.

3.2. Binding of isolated anti-Rha Abs to plant polysaccharides: PGA data

Most of the PS on the plant array (Supplementary Material 2) also contain rhamnose as a part of their backbone. However, only 16 species, in which the rhamnose moiety was located in the pendant substituent linked by α1–4, interacted with human anti-Rha Abs (Fig. 2). The structural features of the interacting plant PS are similar to those described above for bacteria. Considering that plant PS are highly heterogeneous compared to bacterial ones, we present here only two characteristic structures of plant PS (Fig. 2), which are the same (or similar) for several PS in the arrays used [26,27].

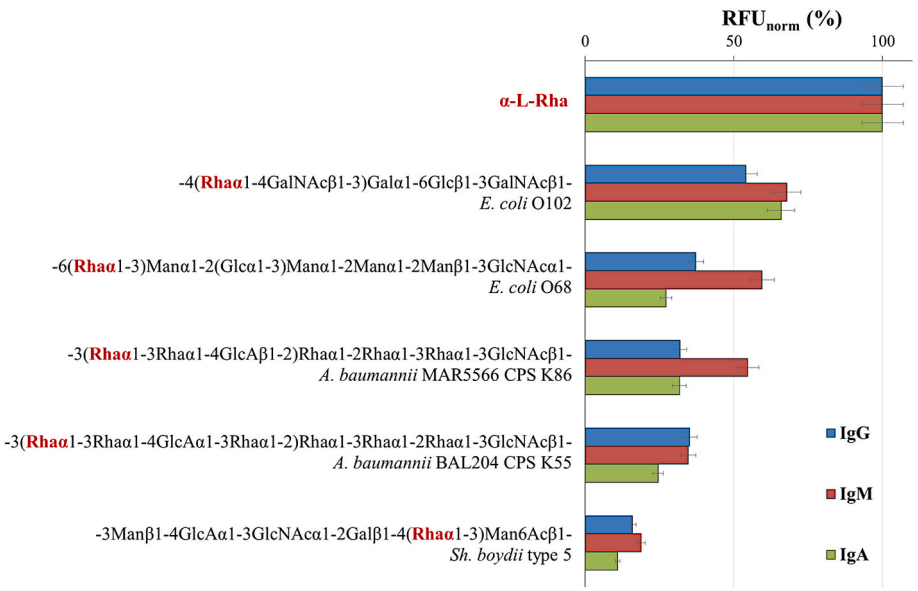


Fig. 1. Five bacterial polysaccharides that have demonstrated the ability to bind human anti-Rha Abs, PGA data. The upper data set corresponds to the α-L-Rha monosaccharide on the same array. The structures of polysaccharides are shown in "glycan form", with monosaccharides considered pyranoses by default, rhamnose is classified as an l-sugar (by default), while all others are D-sugars. All abbreviations were used according to the polysaccharide database at http://csdb.glycoscience.ru/database.

Table 1
Rhamnose-terminated PS, which do not interact with anti-α-L-Rha antibodies. Terminal rhamnose residue is bolded.

PS structure[a]	Source of the PS
-[3(**Rhaα1-4**)GlcAα1-2Rhaα1-2Rhaα1-2Galα1-3GalNAcβ1]-	*E. coli* O120, *O*-PS
-[4(**Rhaα1-3**)GlcNAcα1-2Glcβ1-3Tal2Ac6dα1-3GlcNAcβ1]-	*A. baumannii* LUH5538 K83, capsular PS
-[4(**Rhaα1-2**Fucα1-3)Manα1-3Fucα1-3GlcNAcβ1]-	*E. coli* O36, *O*-PS
-[3(**Rhaα1-4**)GlcNAcα1-2Glcβ1-3Tal6dα1-3GlcNAcβ1]-	*A. baumannii* MAR24 K112, capsular PS
-[4(**Rhaα1-3**)(Glcα1-6)Glcα1-3GalNAcα1-3GalNAcβ1]-	*E. coli* O158, *O*-PS
-[3(**Rhaα1-4**)GlcAβ1-2Rhaα1-2Rhaα1-3Rhaα1-2Rhaα1-3GlcNAcβ1]-	*A. baumannii* LUH5543 K85, capsular PS
-[2**Rhaα1-2**Rhaα1-2Rhaα1-2Glcα1-3GlcNAc6Acα1]-	*E. coli* O19ab, *O*-PS
-[2**Rhaα1-2**Fucα1-2Galβ1-3GalNAcα1-3GlcNAcβ1]-	*E. albertii* O4, lipopolysaccharide

[a] The structures of polysaccharides are presented in "glycan form" instead conventional "polysaccharide form" for the convenience of comparison with oligosaccharides. This means that we omitted designations of ring size (pyranose/furanose) and, in most cases, configuration (D/L). Accordingly, monosaccharides are considered pyranoses by default; fucose, rhamnose, and talose are L-sugars (on default), while all others are D-sugars. Other abbreviations were used according to the polysaccharide database at http://csdb.glycoscience.ru/database.

3.3. Comparison of PS-binding profiles for isolated anti-Rha Abs, initial immunoglobulin preparation, and preparation exhausted at α-L-Rha adsorbent: bacterial polysaccharide PGA data

The consequence of the high structural diversity of rhamnose motifs in PS should be the diversity of rhamnose-dependent epitopes. To determine the contribution of Abs isolated on the α-L-Rha-Sepharose to the total pool of Abs to rhamnose-dependent epitopes, the flow-through material after the affinity column was analyzed using the bacterial PS array (complete data are presented in Supplementary Material 3 and 4). The binding data (for simplicity, only IgM data is shown, results for IgG is presented in Supplementary Material 5) for the material depleted on α-L-Rha-Sepharose (green bars), the native CIP (blue bars), and the affinity-isolated Abs (red bars) are presented in Fig. 3. The comparison demonstrates that the depleted material differs from the original in only a few cases (note *E. coli* O102 and O68).

3.4. Affinity of isolated anti-Rha Abs: EIA data

IC_{50} value of isolated Abs to α-L-Rha was found to be 241 μM in inhibiting the interaction with the polyacrylamide conjugate of α-L-rhamnose by unconjugated monosaccharide in the form of the α-anomer (Fig. 4a). In order to assess whether the IC_{50} value found is high or low, we compared it with the data for antibodies to human blood group A and B antigens, i.e. antibodies known to cause a rapid and powerful immunological reaction during mismatched transfusion. These reference antibodies were isolated in exactly the same way as the anti-rhamnose antibodies under study. The difference is that their antigen is significantly larger, it is a tetrasaccharide (the structure is shown in the figure caption), which suggests a higher affinity of the corresponding antibodies. The difference was not dramatic (taking into account the above-mentioned difference in glycan size), namely, the IC_{50} value for anti-A was five times lower (Fig. 4b).

3.5. Ability of isolated anti-Rha Abs to bind Rha-Rha motifs in oligosaccharide composition: EIA data

The studied antibodies demonstrated the ability to interact with the Rha residue within the disaccharide motifs Rhaα1-2-*Hexose*, Rhaα1-3-*Hexose*, and Rhaα1-4-*Hexose*, located as pendant substituents with respect to the main chain of the PS. To understand whether the aforementioned *Hexose* residue contributes positively to the interaction or serves only as an inert carrier for the Rha residue, we compared the interaction of Abs with the monosaccharide (in both alpha- and beta-forms) and the disaccharides Rhaα1-2Rhaα and Rhaα1-3Rhaα (Fig. 5a). The presented data indicate that the penultimate monosaccharide residue does not affect the interaction of the studied Abs (either positively or negatively) with rhamnose-terminated glycans. Fig. 5b shows what changes and what does not change with depletion of the CIP immunoglobulin preparation on the Rha-Sepharose adsorbent (red curves - before depletion, blue - after), i.e. the level of binding to

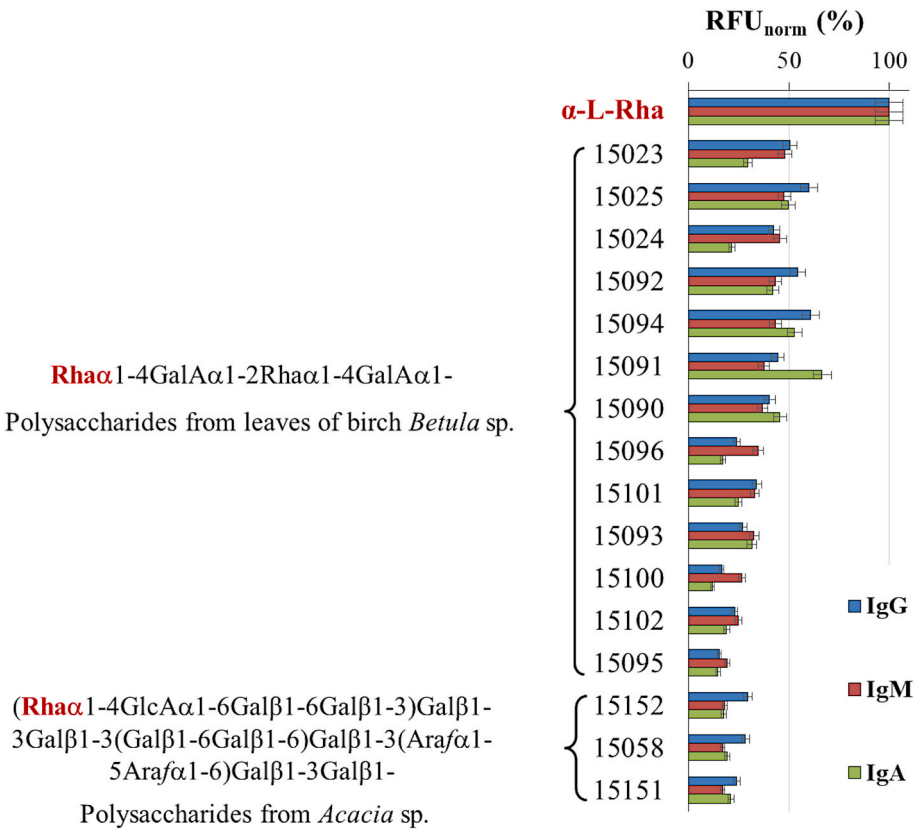


Fig. 2. Plant polysaccharides (selected, see text) that bind isolated human anti-Rha Abs, PGA data. *Betula* sp. from the leaves of birch are pectins with a predominance of rhamnogalacturonan (presented fragment is reflected structure of both a backbone and a side chain of polysaccharides) [26], while *Acacia* sp. are arabinogalactans II [27]. The upper data set corresponds to the α-L-Rha monosaccharide on the same array. The numbers represent are plant PS IDs, with structural descriptions given in Supplementary Material 2. The typical fragments of mentioned plant PS are shown in "glycan form", with monosaccharides considered pyranoses by default (Ara$_f$ is arabinofuranose); rhamnose and arabinose are classified as L-sugar, while all others are D-sugars. Here and elsewhere, RFU$_{norm}$ (%) is fluorescent intensities normalized to 100 %.

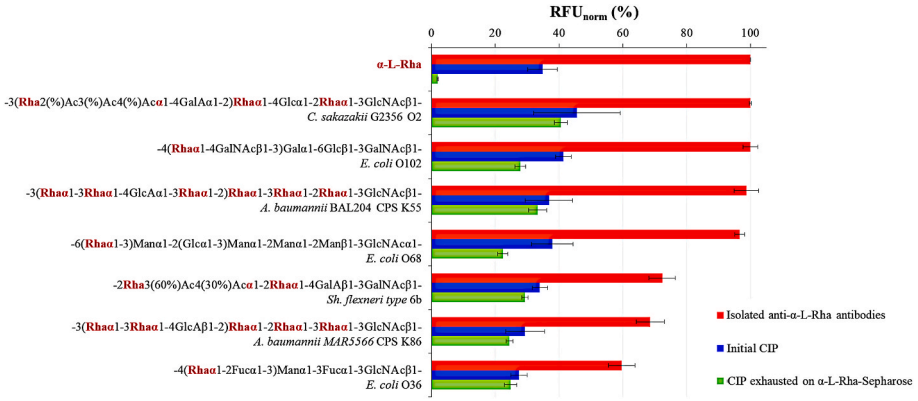


Fig. 3. Binding of IgM Abs to bacterial polysaccharides, PGA data. Affinity-isolated Abs are compared with the native unprocessed CIP, as well as the material that did not bind to the αRha-Sepharose resin. The accompanying table shows the repeating unit of a PS, with the terminal or pendant rhamnose fragments highlighted in red.

those glycans that are shown in Fig. 5a. This experiment was performed twice with identical results and makes it possible to approximately estimate the proportion of antibodies to the rhamnose monosaccharide among other anti-rhamnose (directed in particular to disaccharide epitopes). The data presented indicate a sharp decrease in the level of anti-monosaccharide Abs, but not anti-disaccharide Abs, that is, among the total antibodies of human blood there are varieties that recognize the motifs Rhaα1-2Rhaα and Rhaα1-3Rhaα, and many other rhamnose-containing ones can be assumed.

4. Discussion

Previous studies have shown that the accuracy of epitope mapping is improved by isolating a narrow pool of Abs from serum or CIP using hapten-specific affinity chromatography [3]. For instance, when analyzing the specificity of human anti-Tn antibodies using the GalNAcα-Sepharose adsorbent, immunoglobulins were isolated that bind not only to the monosaccharide, but also to GalNAcα-containing oligosaccharides and glycopeptides [28]. Moreover, the interaction of

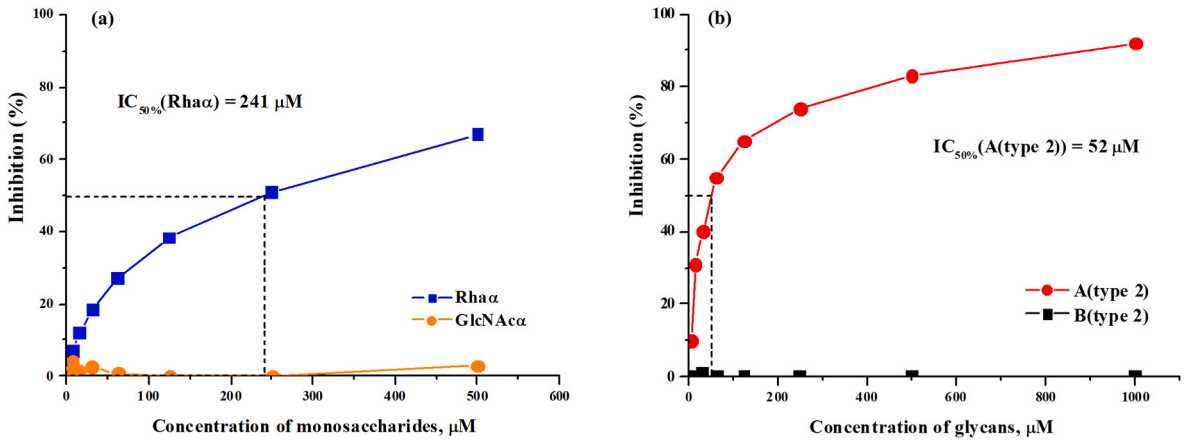


Fig. 4. (a) Inhibition of the interaction between anti-α-L-Rha Abs and αRha-PAA coated on an EIA microplate using unconjugated monosaccharide L-Rhaα-spacer (spacer = $OCH_2CH_2CH_2NH_2$). The concentration of analyzed anti-α-L-Rha antibodies (IgG + IgM + IgA) was 0.2 μg/mL. Unconjugated monosaccharide GlcNAcα-spacer was used as a negative control; (b) For comparison, inhibition of interaction of affinity isolated Abs to blood group A(type 2) tetrasaccharide with A(type 2)-PAA on an EIA microplate with unconjugated A(type 2) tetrasaccharide. A(type 2) is GalNAcα1-3(Fucα1-2)Galβ1-4GlcNAcβ. B(type 2) taken as negative control is Galα1-3(Fucα1-2)Galβ1-4GlcNAcβ.

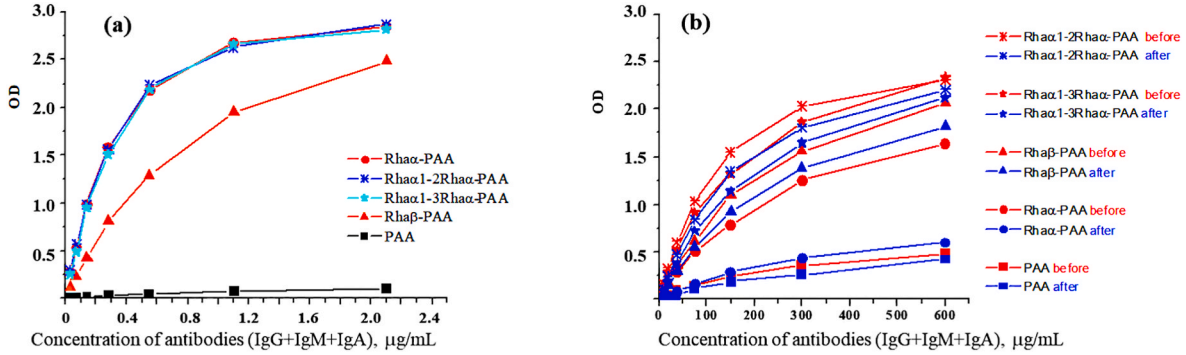


Fig. 5. (a) Binding of isolated anti-α-L-Rha Abs to the monosaccharides α-L-Rha, and β-l-Rha, as well as disaccharides Rhaα1-2Rhaα and Rhaα1-2Rhaα. All four glycans are in the form of PAA conjugates, and equipped with the same spacer. (b) Binding of native CIP ("before") compared to CIP exhausted on α-L-Rha-Sepharose ("after") against the same set of glycans. EIA data.

isolated Abs with the glycan used as the affinity ligand is sometimes noticeably weaker than with other oligosaccharides. This affinity purification methodology was used in this study to isolate anti-α-L-Rha Abs on α-L-Rha-Sepharose adsorbent. These purified Abs were then analyzed for specificity against ~1000 glycans.

Ultimately, the objective of this study of human anti-Rha Abs was to answer or gain further insight into the question: are these immunoglobulins naturally occurring or adaptive? If naturally occurring, what are they primarily targeted against, and what causes the priming of the corresponding B1 lymphocytes? If adaptive, then what caused the immune response? Does their recognition of plant polysaccharides have a physiological basis, or is it just a consequence of some similarity between plant and bacterial PS?

Current knowledge does not allow for the discrimination of natural Abs (nAbs, the product of B1 lymphocytes [29]) from adaptive immunoglobulins based on the structure of the polypeptide chain or its post-translational modification, so, we can only speculate based on indirect data. Since the lipopolysaccharide *O*-antigens of various gram-negative bacteria and polysaccharides of gram-positive infectious bacteria contain α-L-rhamnose, therapeutic immunoglobulins (from which the Abs in this study were isolated, and which are typically produced from 1000 or more donor plasmas) must contain adaptive Abs. However, it is not known how many individual donations (ranging from a few to many hundreds) within the pool actually contributed to the anti-α-L-Rha Abs that we isolated by hapten-specific affinity chromatography. In fact, a single donation with a very high level of Abs may be the dominant source of specific Abs (as often seen with ABO antibodies in intravenous immunoglobulin, IVIG). However, profiling of individual blood samples shows the presence of anti-Rha Abs in all individuals [8,19] at high or moderate titers. It is very unlikely that all donors had a recent history of infection, suggesting that at least a significant part of the affinity-isolated anti-α-L-Rha Abs are nAbs and, therefore, important for maintaining anti-bacterial innate immunity.

Due to the small size of the recognizable epitope (just a monosaccharide), one would expect a low affinity of Abs at the level of a single antibody binding fragment. Considering the high proportion of IgM and IgA, that a priori interact polyvalently, we assume, that the isolated immunoglobulins bind several Rha residues simultaneously, which significantly increases their affinity to the target [30,31]. Furthermore, since the natural targets of anti-α-L-Rha immunoglobulins are lipopolysaccharides tightly packed in the bacterial cell wall (as well as capsular PS), we can assume that they recognize molecular patterns (PS bundles) rather than several individual epitopes on a single PS chain. This pattern recognition is supported by the unexpected property that anti-α-L-Rha Abs (isolated on Sepharose with an immobilized α-anomer) are not able to distinguish between the α- and β-anomers of L-rhamnose, as evidenced by the following experiments: 1) according to the PGA data (Supplementary Material 1 and 2), the β-anomer, although inferior to the α-anomer, shows a moderate level of binding; 2) in EIA, both α-L-Rha-PAA and β-l-Rha-PAA are active in direct binding (Fig. 5a); 3)

two polysaccharides with terminal β-L-rhamnose residues, namely -[3Rhaβ1-4(Glcα1-3)Rhaα1-2Rhaα1-3Galα1-3-D-FucNAcα1]- from *E. cloacae* G3421, and -[3Rhaβ1-4Rhaα1-2Rhaα1-2GalAα1-3GalNAcα1]- from *Sh. boydii* type 18 (see their detailed structural formulas in Supplementary Material 6 and in http://csdb.glycoscience.ru/database), were found to weakly bind Abs in the bacterial PS array (Supplementary Material 3). The absence of dependence on the anomeric configuration of the rhamnose residue significantly supports the ability of several antigen-binding fragments of the same immunoglobulin to bind an arrangement consisting of multiple rhamnoses, i.e. a molecular pattern. There are also arguments suggesting that the epitope recognized by anti-α-L-Rha antibodies is more complex, than just the monosaccharide; in addition to rhamnose, their paratope may recognize other regions of the PS chain. In this case, one should not expect monospecific homogeneous Abs, but rather a set of Abs that differ significantly in fine specificity. Indeed, this is indirectly indicated by the observed diversity of PS structures capable of binding to seemingly anti-monosaccharide Abs.

The IC_{50} value found (Fig. 4) for inhibition of anti-Rha Abs by α-rhamnose monosaccharide is in the same range as that for inhibition of other human blood Abs to monosaccharides (for example, GalNAcα [28]) isolated in a similar manner. The IC_{50} is ~4–5 times higher (indicating lower affinity) than for nAbs to larger antigens, such as the Galili trisaccharide or the blood group A tetrasaccharide (Fig. 4b). However, these Abs are unable to discriminate monosaccharide Rha from rhamno-disaccharides (Fig. 5a). Thus, we have arguments supporting both the recognition of a monosaccharide epitope by the studied affinity-isolated Abs and the possibility of a more extended epitope. It is impossible to exclude the validity of both points of view; that is, the polyclonal Abs studied are a combination of pattern-recognizing nAbs (where the epitope is just monosaccharide Rha) and oligosaccharide-specific adaptive Abs. This idea is in agreement with the data on the specificity pattern of the antibody pool depleted on the monosaccharide adsorbent (Fig. 5b); indeed, the exhausting procedure removes antibodies capable of recognizing the pendant Rha monosaccharide regardless of the structure of the polysaccharide backbone, and does not remove those that are directed to a more complex epitope.

The isotype composition IgG : IgM : IgA = 1.0 : 3.3 : 2.8, indicating a prevalence of IgM content, supports the notion of nAbs. A retrospective analysis of accumulated PGA data (not shown) from individual donors indicates that the level of anti-Rha IgG is sometimes low, while the level of IgM is consistently high or moderate/closer to high, suggesting that these Abs most likely belong to nAbs.

Priming of B1 cells producing anti-glycan nAbs is thought to be mediated by microbiota [3]. We believe that this also applies to anti-Rha Abs, especially considering the high occurrence of rhamno-PS in bacteria. Although the structure of PS in non-infectious bacteria has been studied much less than in infectious bacteria ([6], http://csdb.glycoscience.ru/database), the repertoire of corresponding Abs is quite broad [32], even in the first months of life, when innate immunity has started to form. This suggests that intestinal bacteria are likely key factors of priming Rha-specific B1 cells.

Another factor contributing to the emergence of adaptive anti-Rha Abs in a significant number of people may the action of polysaccharide-based broadly protective vaccines, particularly against *Shigella flexneri*, whose PS contain a terminal rhamnose residue [33]. However, other molecules should not be excluded, particularly, rhamno-lipids (where the glycan part consist of mono- or di-rhamnosides) produced by *Pseudomonas aeruginosa* [34], which can also be found in some cosmetic preparations. Additionally, rhamnolipids, α-solanine, and α-chaconine – potato alkaloids with a rhamnolipid-containing trisaccharide moiety – should also be noted. Based on this assumption, we tested potato sprout extract and observed inhibition of the anti-Rha Abs (Fig. 6).

This study demonstrated the binding of human anti-Rha Abs to numerous plant PS, which requires further explanation. The repertoire

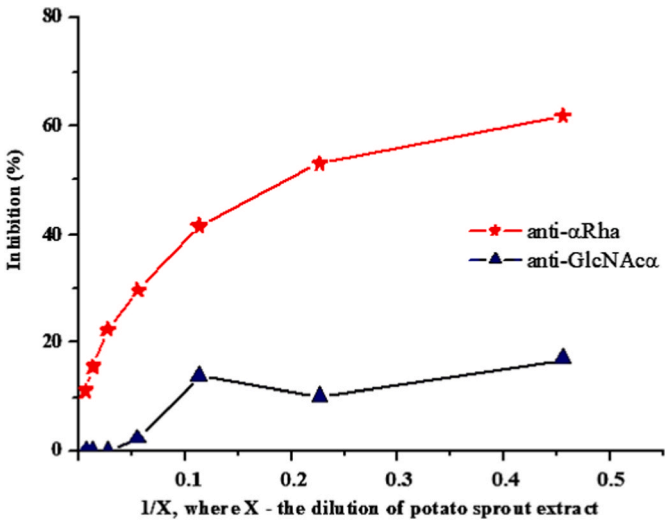


Fig. 6. A component of potato sprout extract specifically blocks interaction of anti-α-L-Rha antibodies with α-L-rhamnopyranose. The data of inhibitory EIA are shown. Glycoconjugates α-L-Rha-PAA and GlcNAcα-PAA were immobilized on an EIA plate for analysis of anti-α-L-Rha and anti-GlcNAcα (negative control) antibodies, respectively.

of human Abs to plant PS is by no means limited to anti-Rha Abs, and the range is quite diverse [25,35]. From this perspective, the interaction of anti-Rha Abs (naturally targeted at bacterial PS) with plant PS is to be expected.

For bacterial PS, there is an extremely large diversity of structural motifs of the *Rha-Hex-Hex-Hex* type. Therefore, for innate immunity, it is "more advantageous" to recognize the rhamnose monosaccharide rather than a variety of specific rhamno-oligosaccharide motifs. The data presented here supports this idea, as the pool of "anti-Rha" Abs contained immunoglobulins that didn't discriminate among the *-XYZ* fragments in composition of *Rha-XYZ*. However, it is important to consider their content in the pool and whether they are dominant. To investigate this, Abs capable of binding to the monosaccharide were exhaustively isolated from the total pool of immunoglobulins using an excess of the Rha-adsorbent. The remaining material was then examined for binding to rhamno-PS of bacterial origin (Fig. 3 and Supplementary Material 3 and 4). The results indicate that the binding patterns of the original CIP and the CIP depleted with the rhamnose adsorbent to rhamno-PS are virtually identical. We suggest that any PS has several antigenic determinants that can be recognized by nAbs. Under the conditions of the described experiments, only Abs against one of these determinants were removed, which did not affect the overall pattern of interaction with the PS array. Therefore, this approach did not answer the question of the quantitative proportion of anti-monosaccharide Abs in the population of Abs against rhamno-containing antigenic determinants. Nevertheless, the concentration of the Abs in question in blood, particularly in relation to the content of other major anti-glycan Abs, indirectly supports a high degree of representation of Abs to the rhamnose monosaccharide among nAbs, and therefore highlights their significant role.

5. Conclusion

The human antibodies known as "anti-rhamnose" require more attention than they have received so far. They are present in all healthy individuals tested without exception, which has not been observed for other antibodies, even for the well-known ones to the xenoantigen "αGal", Galα1-3Galβ1-4GlcNAc. Their activity (sometimes the term "titer" is used), i.e. the parameter proportional to affinity multiplied by content in the blood, turned out to be one of the highest among human nAbs, comparable with the above-mentioned antibodies to the

xenoantigen "αGal" and to the blood group A and B antigens. This gives grounds to believe that antibodies to Rha are vitally important, apparently performing the function of eliminating infectious bacteria, a significant cohort of which have lipopolysaccharides or capsular polysaccharides decorated with a rhamnose residue. The high proportion of multipoint-reactive IgM and IgA suggests an attack primarily on lipopolysaccharides, which form a high-density antigen pattern on the bacterial cell wall. For the above reasons, anti-rhamnose antibodies are particularly attractive in cancer treatment strategies in which tumor cells are modified with a foreign antigen to which the patient has pre-existing antibodies, initiating a further, comprehensive immune response.

CRediT authorship contribution statement

Nadezhda V. Shilova: Writing – review & editing, Methodology, Investigation, Conceptualization. **Polina S. Obukhova:** Writing – review & editing, Methodology, Investigation. **Yuriy A. Knirel:** Methodology, Investigation. **Victoria V. Golovchenko:** Resources, Methodology, Investigation. **Olga A. Patova:** Resources, Methodology, Investigation. **Svetlana V. Tsygankova:** Writing – review & editing, Methodology, Investigation. **Polina V. Mikshina:** Validation, Resources, Investigation. **Stephen Henry:** Writing – review & editing, Investigation, Conceptualization. **Nicolai V. Bovin:** Writing – original draft, Project administration, Investigation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.carres.2025.109717.

Data availability

Data will be made available on request.

References

[1] H. Ahmad, E.K. Chapnick, Conjugated polysaccharide vaccines, Infect. Dis. Clin. 13 (1999) 113–133, https://doi.org/10.1016/S0891-5520(05)70046-5.
[2] M. Ghattas, G. Dwivedi, M. Lavertu, M.-G. Alameh, Vaccine technologies and platforms for infectious diseases: current progress, challenges, and opportunities, Vaccines 9 (2021) 1490, https://doi.org/10.3390/vaccines9121490.
[3] N. Bovin, P. Obukhova, O. Galanina, N. Antipova, K. Dobrochaeva, N. Khasbiullina, N. Shilova, Is it time to switch over to glyco molecular patterns? in: D. Banerjee (Ed.), Glycome Hidden Code Biol Nova Scien, 2021, pp. 377–395.
[4] S. Panda, J.L. Ding, Natural antibodies bridge innate and adaptive immunity, J. Immunol. 194 (2015) 13–20, https://doi.org/10.4049/jimmunol.1400844.
[5] K. Kappler, T. Hennet, Emergence and significance of carbohydrate-specific antibodies, Gene Immun. 21 (2020) 224–239, https://doi.org/10.1038/s41435-020-0105-9.
[6] S.R. Stowell, C.M. Arthur, R. McBride, O. Berger, N. Razi, J. Heimburg-Molinaro, L. C. Rodrigues, J.-P. Gourdine, A.J. Noll, S. von Gunten, D.F. Smith, Y.A. Knirel, J. C. Paulson, R.D. Cummings, Microbial glycan microarrays define key features of host-microbial interactions, Nat. Chem. Biol. 10 (2014) 470–476, https://doi.org/10.1038/nchembio.1525.
[7] R.N.D. Luetscher, T.R. McKitrick, C. Gao, A.Y. Mehta, A.M. McQuillan, R. Kardish, K.F. Boligan, X. Song, L. Lu, J. Heimburg-Molinaro, S. von Gunten, G. Alter, R. D. Cummings, Unique repertoire of anti-carbohydrate antibodies in individual human serum, Sci. Rep. 10 (2020) 15436, https://doi.org/10.1038/s41598-020-71967-y.
[8] M.E. Huflejt, M. Vuskovic, D. Vasiliu, H. Xu, P. Obukhova, N. Shilova, A. Tuzikov, O. Galanina, B. Arun, K. Lu, N. Bovin, Anti-carbohydrate antibodies of normal sera: findings, surprises and challenges, Mol. Immunol. 46 (2009) 3037–3049, https://doi.org/10.1016/j.molimm.2009.06.010.
[9] O. Oyelaran, L.M. McShane, L. Dodd, J.C. Gildersleeve, Profiling human serum antibodies with a carbohydrate Antigen microarray, J. Proteome Res. 8 (2009) 4301–4310, https://doi.org/10.1021/pr900515y.
[10] S. Li, F. Chen, Y. Li, L. Wang, H. Li, G. Gu, E. Li, Rhamnose-containing compounds: biosynthesis and applications, Molecules 27 (2022) 5315, https://doi.org/10.3390/molecules27165315.
[11] A. Vanacore, G. Vitiello, A. Wanke, D. Cavasso, L.A. Clifton, L. Mahdi, M. A. Campanero-Rhodes, D. Solís, M. Wuhrer, S. Nicolardi, A. Molinaro, R. Marchetti, A. Zuccaro, L. Paduano, A. Silipo, Lipopolysaccharide O-antigen molecular and supramolecular modifications of plant root microbiota are pivotal for host recognition, Carbohydr. Polym. 277 (2022) 118839, https://doi.org/10.1016/j.carbpol.2021.118839.
[12] P. Concórdio-Reis, V.D. Alves, X. Moppert, J. Guézennec, F. Freitas, M.A.M. Reis, Characterization and biotechnological potential of extracellular polysaccharides synthesized by alteromonas strains isolated from French Polynesia marine environments, Mar. Drugs 19 (2021) 522, https://doi.org/10.3390/md19090522.
[13] P. Garcia-Vello, G. Sharma, I. Speciale, A. Molinaro, S.-H. Im, C. De Castro, Structural features and immunological perception of the cell surface glycans of Lactobacillus plantarum: a novel rhamnose-rich polysaccharide and teichoic acids, Carbohydr. Polym. 233 (2020) 115857, https://doi.org/10.1016/j.carbpol.2020.115857.
[14] N. Jiang, F.M. Dillon, A. Silva, L. Gomez-Cano, E. Grotewold, Rhamnose in plants - from biosynthesis to diverse functions, Plant Sci. 302 (2021) 110687, https://doi.org/10.1016/j.plantsci.2020.110687.
[15] M.-Y. Mistou, I.C. Sutcliffe, N.M. van Sorge, Bacterial glycobiology: rhamnose-containing cell wall polysaccharides in Gram-positive bacteria, FEMS Microbiol. Rev. 40 (2016) 464–479, https://doi.org/10.1093/femsre/fuw006.
[16] C. De Castro, A. Molinaro, F. Piacente, J.R. Gurnon, L. Sturiale, A. Palmigiano, R. Lanzetta, M. Parrilli, D. Garozzo, M.G. Tonetti, J.L. Van Etten, Structure of N-linked oligosaccharides attached to chlorovirus PBCV-1 major capsid protein reveals unusual class of complex N-glycans, Proc. Natl. Acad. Sci. 110 (2013) 13956–13960, https://doi.org/10.1073/pnas.1313005110.
[17] V. Martinez, M. Ingwers, J. Smith, J. Glushka, T. Yang, M. Bar-Peled, Biosynthesis of UDP-4-keto-6-deoxyglucose and UDP-rhamnose in Pathogenic Fungi magnaporthe grisea and Botryotinia fuckeliana, J. Biol. Chem. 287 (2012) 879–892, https://doi.org/10.1074/jbc.M111.287367.
[18] Y. Yu, Q. Wen, A. Song, Y. Liu, F. Wang, B. Jiang, Isolation and immune activity of a new acidic Cordyceps militaris exopolysaccharide, Int. J. Biol. Macromol. 194 (2022) 706–714, https://doi.org/10.1016/j.ijbiomac.2021.11.115.
[19] R.T.C. Sheridan, J. Hudon, J.A. Hank, P.M. Sondel, L.L. Kiessling, Rhamnose glycoconjugates for the recruitment of endogenous anti-carbohydrate antibodies to tumor cells, Chembiochem 15 (2014) 1393–1398, https://doi.org/10.1002/cbic.201402019.
[20] H. Hong, K. Zhou, H. Lin, Y. Li, Z. Wu, In situ self-assembly of antibody-rhamnose complex as a pre-targeting strategy for enhanced cancer immunotherapy, Chem. Eur J. 31 (2025), https://doi.org/10.1002/chem.202403539.
[21] C. Ou, S.K. Prabhu, X. Zhang, G. Zong, Q. Yang, L. Wang, Synthetic antibody-rhamnose cluster conjugates show potent complement-dependent cell killing by recruiting natural antibodies, Chem. Eur J. 28 (2022), https://doi.org/10.1002/chem.202200146.
[22] S.C.S.C. Churms, E.H.E.H. Merrifield, A.M. Stephen, S. A.M., A.M. Stephen, Some new aspects of the molecular structure of Acacia Senegal gum (gum arabic), Carbohydr. Res. 123 (1983) 267–279, https://doi.org/10.1016/0008-6215(83)88483-3.
[23] G.V. Pazynina, S.V. Tsygankova, P.S. Obukhova, N.V. Shilova, A.S. Paramonov, A. O. Chizhov, Y.A. Knirel, N.V. Bovin, Synthesis of disaccharides for the study of human blood antibodies capable of recognizing the inner Glcβ1-3GalNAc disaccharide fragment of bacterial polysaccharides, Mendeleev Commun. 33 (2023) 476–478, https://doi.org/10.1016/j.mencom.2023.06.011.
[24] P. Obukhova, S. Tsygankova, A. Chinarev, N. Shilova, A. Nokel, P. Kosma, N. Bovin, Are there specific antibodies against Neu5Gc epitopes in the blood of healthy individuals? Glycobiology 30 (2020) 395–406, https://doi.org/10.1093/glycob/cwz107.
[25] A.V. Nikiforova, V.V. Golovchenko, P.V. Mikshina, O.A. Patova, T.A. Gorshkova, N. V. Bovin, N.V. Shilova, Plant polysaccharide array for studying carbohydrate-binding proteins, Biochem. 87 (2022) 890–902, https://doi.org/10.1134/S0006297922090036.
[26] V.V. Golovchenko, V.A. Khlopin, O.A. Patova, L.S. Feltsinger, M.I. Bilan, A. S. Dmitrenok, A.S. Shashkov, Pectin from leaves of birch (Betula pendula Roth.): results of NMR experiments and hypothesis of the RG-I structure, Carbohydr. Polym. 284 (2022) 119186, https://doi.org/10.1016/j.carbpol.2022.119186.
[27] A.E. Clarke, R.L. Anderson, B.A. Stone, Form and function of arabinogalactans and arabinogalactan-proteins, Phytochemistry 18 (1979) 521–540, https://doi.org/10.1016/S0031-9422(00)84255-7.

[28] K. Dobrochaeva, N. Khasbiullina, N. Shilova, N. Antipova, P. Obukhova, T. Ovchinnikova, O. Galanina, O. Blixt, H. Kunz, A. Filatov, Y. Knirel, J. LePendu, S. Khaidukov, N. Bovin, Specificity of human natural antibodies referred to as anti-Tn, Mol. Immunol. 120 (2020) 74–82, https://doi.org/10.1016/j.molimm.2020.02.005.

[29] Y.S. Choi, J.A. Dieter, K. Rothaeusler, Z. Luo, N. Baumgarth, B-1 cells in the bone marrow are a significant source of natural IgM, Eur. J. Immunol. 42 (2012) 120–129, https://doi.org/10.1002/eji.201141890.

[30] D.W. Mason, A.F. Williams, The kinetics of antibody binding to membrane antigens in solution and at the cell surface, Biochem. J. 187 (1980) 1–20, https://doi.org/10.1042/bj1870001.

[31] D.M. Czajkowsky, Z. Shao, The human IgM pentamer is a mushroom-shaped molecule with a flexural bias, Proc. Natl. Acad. Sci. 106 (2009) 14960–14965, https://doi.org/10.1073/pnas.0903805106.

[32] N.R. Khasbiullina, N.V. Shilova, M.J. Navakouski, A.Y. Nokel, O. Blixt, L. O. Kononov, Y.A. Knirel, N.V. Bovin, The repertoire of human antiglycan antibodies and its dynamics in the first year of life, Biochem 84 (2019), https://doi.org/10.1134/S0006297919060038.

[33] G. Desalegn, N. Kapoor, L. Pill-Pepe, L. Bautista, L. Yin, E. Ndungo, E.V. Oaks, J. Fairman, M.F. Pasetti, A novel shigella O-Polysaccharide–IpaB conjugate vaccine elicits robust antibody responses and confers protection against multiple shigella serotypes, mSphere 8 (2023), https://doi.org/10.1128/msphere.00019-23.

[34] U.A. Ochsner, A.K. Koch, A. Fiechter, J. Reiser, Isolation and characterization of a regulatory gene affecting rhamnolipid biosurfactant synthesis in Pseudomonas aeruginosa, J. Bacteriol. 176 (1994) 2044–2054, https://doi.org/10.1128/jb.176.7.2044-2054.1994.

[35] H. Kiyohara, T. Matsumoto, T. Nagai, S.-J. Kim, H. Yamada, The presence of natural human antibodies reactive against pharmacologically active pectic polysaccharides from herbal medicines, Phytomedicine 13 (2006) 494–500, https://doi.org/10.1016/j.phymed.2005.09.004.